

EXPERIMENT 10:

```
Print("THASLIMA NASREEN/192424152")

import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.mixture import GaussianMixture
from sklearn.preprocessing import StandardScaler

# Load dataset
iris = load_iris()
X = iris.data

# Select two features for visualization
X = X[:, [0, 2]]

# Standardize the data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Create Gaussian Mixture Model
gmm = GaussianMixture(
    n_components=3,
    covariance_type='full',
    random_state=42
)

# Train the model using EM algorithm
gmm.fit(X_scaled)

# Predict cluster labels
labels = gmm.predict(X_scaled)

# Display cluster labels
print("Cluster Labels:")
print(labels)
```

```

# Display cluster means

print("\nCluster Means:")
print(gmm.means_)

# Display covariance matrices
print("\nCovariance Matrices:")
print(gmm.covariances_)

# Calculate log-likelihood
log_likelihood = gmm.score(X_scaled)
print("\nLog-Likelihood:", log_likelihood)

# Display number of iterations
print("Number of EM Iterations:", gmm.n_iter_)

# Display cluster probabilities
print("\nCluster Probabilities for First 5 Samples:")
print(gmm.predict_proba(X_scaled)[:5])

# Plot the clusters
plt.figure(figsize=(8, 6))
plt.scatter(
    X_scaled[:, 0],
    X_scaled[:, 1],
    c=labels,
    cmap="viridis",
    s=50
)
plt.scatter(
    gmm.means_[:, 0],
    gmm.means_[:, 1],
    marker="X",
    s=200,

```

```
color="red",
label="Cluster Centers"
)

plt.xlabel("Sepal Length (Standardized)")
plt.ylabel("Petal Length (Standardized)")
plt.title("Expectation Maximization - Gaussian Mixture Model")
plt.legend()
plt.show()
```

OUTPUT:

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Cluster Labels:

```
[111111111111111111111111111111111111111111111  
1111111111111000202220222222022222222  
00002222222202222222222222222022020000  
2022220020202002220002220222000200020  
22]
```

Cluster Means:

$$\begin{bmatrix} 1.06161914 & 0.889505 \\ -1.01454279 & -1.30502935 \\ 0.10476089 & 0.48023932 \end{bmatrix}$$

Covariance Matrices:

```
[[[0.52516547 0.27889389]
 [0.27889389 0.20972483]]
 [[0.17879104 0.01104849]
 [0.01104849 0.0095402 ]]
 [[0.33623677 0.16674602]
 [0.16674602 0.15395216]]]
```

Log-Likelihood: -1.345914450018518

Number of EM Iterations: 8

Cluster Probabilities for First 5 Samples:

```
[[5.83774336e-08 9.99999582e-01 3.60112754e-07]
[2.22038027e-07 9.99998499e-01 1.27862049e-06]
[4.40885113e-07 9.99997184e-01 2.37559889e-06]
[3.94617198e-06 9.99975025e-01 2.10284587e-05]
[1.14233472e-07 9.99999205e-01 6.80563853e-07]]
```

